

## SCREENING TEST - JUNIOR LEVEL - 1998

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1. The value of  $\left(\frac{1999}{6}\right)^2 - \left(\frac{1997}{6}\right)^2$  is
- A)  $\frac{1}{9}$       B)  $\frac{2}{3}$       C) 111      D) 222      E) 1332
2. A man drives from his house to the station. If he drives at the rate of 10 kms per hour, he reaches the station at 6 pm. If he drives at 15 kms per hour, he would reach the station at 4 pm. At what speed, in kms per hour, should he drive so as to reach the station at 5 pm?
- A) 12      B)  $12\frac{1}{3}$       C)  $5\sqrt{5}$       D)  $12\frac{1}{2}$       E) 13
3. If  $n$  men produce  $n$  samples of a product by working  $n$  hours a day for  $n$  days, the number of samples produced by  $m$  men working for  $m$  hours a day for  $m$  days is
- A)  $\frac{n^3}{m^2}$       B)  $\frac{m^3}{n^2}$       C)  $\frac{n^2}{m^3}$       D)  $\frac{m^2}{n^3}$       E)  $m$
4. In a particular year, the month of January had exactly four Thursdays and four Sundays. On which day of the week did January 1st occur that year?
- A) Monday      B) Tuesday      C) Wednesday  
D) Thursday      E) Friday
5. Ann, Bert and Chris each wear a jeans of different colours (blue, black or grey). It is known that if Ann wears blue then Bert wears grey ; if Ann wears black Chris wears grey and if Bert doesn't wear black, Chris wears blue. The jeans Ann is wearing has the colour
- A) blue      B) black      C) grey      D) which cannot be determined
6. If  $(3x-1)^7 = a_7x^7 + a_6x^6 + \dots + a_0$ , then  $a_7 + a_6 + \dots + a_0$  is

What is the smallest positive integer  $n$  such that  $\sqrt{n} - \sqrt{n-1} < 0.01$ ?

- A) 2499      B) 2500      C) 2501      D) 10000      E) 10001

If  $3x^3 - 9x^2 + kx - 12$  is divisible by  $x - 3$ , then it is also divisible by

- A)  $3x^2 - x + 4$       B)  $3x^2 - 4$       C)  $3x^2 + 4$       D)  $3x - 4$   
E)  $3x + 4$

If 78 is divided into three parts which are proportional to  $1, \frac{1}{3}, \frac{1}{6}$ , the middle part is

- A)  $9\frac{1}{3}$       B) 13      C)  $17\frac{1}{3}$       D)  $18\frac{1}{3}$       E) 26

0. Find the smallest positive number from the numbers below.

- A)  $10 - 3\sqrt{11}$       B)  $3\sqrt{11} - 10$       C)  $18 - 5\sqrt{13}$   
D)  $51 - 10\sqrt{26}$       E)  $10\sqrt{26} - 51$

1. In a farm both men and women are working. Exactly one-third of the staff bring one child each. One day each male employee planted 13 trees, each women employee planted 10 trees and each child planted 6 trees. A total of 159 trees were planted on that day. How many women employees are there?

- A) 2      B) 5      C) 7      D) 12  
E) cannot be determined from given information.

2.  $\frac{2\sqrt{6}}{\sqrt{2} + \sqrt{3} + \sqrt{5}}$  equals

- A)  $\sqrt{2} + \sqrt{3} - \sqrt{5}$       B)  $4 - \sqrt{2} - \sqrt{3}$       C)  $\sqrt{2} + \sqrt{3} + \sqrt{6} - 5$   
D)  $\frac{1}{2}(\sqrt{2} + \sqrt{5} - \sqrt{3})$       E)  $\frac{1}{3}(\sqrt{3} + \sqrt{5} - \sqrt{2})$

3. The largest integer  $n$  for which  $n^{200} < 5^{300}$  is

- A) 8      B) 9      C) 10      D) 11      E) 12

4.  $\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}}$  is equal to

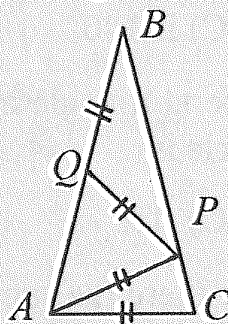


15. For a real number  $x$  and a positive integer  $n$ , define

$$\binom{x}{n} = \frac{x(x-1)(x-2) \dots (x-(n-1))}{1.2.3 \dots n}.$$

What is  $\binom{-\frac{1}{2}}{100} \div \binom{\frac{1}{2}}{100}$ ?

- A) -199      B) -197      C) -1      D) 197      E) 199
16. The units digit of  $3^{1001} \times 7^{1002} \times 13^{1003}$  is
- A) 1      B) 3      C) 5      D) 7      E) 9
17. The sides of a right angled triangle are  $a$ ,  $a+d$  and  $a+2d$ , with  $a$  and  $d$  both positive. The ratio of  $a$  to  $d$  is
- A) 1:3      B) 1:4      C) 2:1      D) 3:1      E) 3:4
18. Each edge of a cube is increased by 50%. The percent of increase in the surface area of the cube is
- A) 50      B) 125      C) 150      D) 300      E) 750
19. Triangle  $ABC$  is isosceles with base  $AC$ . Points  $P$  and  $Q$  are respectively in  $CB$  and  $AB$  such that  $AC = AP = PQ = QB$ . Angle  $B$  in degrees is



- A)  $25\frac{5}{7}$       B)  $26\frac{1}{3}$       C) 30      D) 40      E) not determined by information given.
20. In racing over a given distance  $d$  at uniform speed,  $A$  can beat  $B$  by 30 meters,  $B$  can beat  $C$  by 20 meters and  $A$  can beat  $C$  by 48 meters, then  $d$ , in meters, equals
- A) 300      B) 600      C) 900      D) 100      E) 1050
21. Let  $x = 0.123456789101112 \dots 998999$ , where the digits are obtained by writing the integers 1 through 999 in order. The 1983<sup>rd</sup> digit to the right of the decimal point is

22. If  $N = \frac{\sqrt{\sqrt{5}+2} + \sqrt{\sqrt{5}-2}}{\sqrt{\sqrt{5}+1}} - \sqrt{3-2\sqrt{2}}$ , then  $N$  equals

- A) 1      B)  $2\sqrt{2} - 1$       C)  $\frac{\sqrt{5}}{2}$       D)  $\sqrt{\frac{5}{2}}$       E)  $\sqrt{5} - 1$

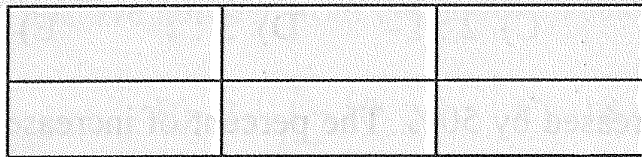
23. A circle is inscribed in an equilateral triangle and a square is inscribed in the circle. The ratio of the area of the triangle to the area of the square is

- A)  $\sqrt{3}:1$       B)  $\sqrt{3}:\sqrt{2}$       C)  $3\sqrt{3}:2$       D)  $3:\sqrt{2}$       E)  $3:2\sqrt{2}$

24. The smallest positive integer  $x$  for which  $1260x = N^3$  where  $N$  is an integer is

- A) 1050      B) 1260      C)  $1260^2$       D) 7350      E) 44100

25. In the figure given below the number of ways we can make out rectangles is



- A) 6      B) 7      C) 12      D) 18      E) 9

26. In  $\triangle ABC$ ,  $AB' = AC$  and  $\angle BAC = 40^\circ$ . If  $O$  is such that  $\angle OBC = \angle OCA$ , then the measure of  $\angle BOC$

- A) is  $110^\circ$       B) is  $80^\circ$       C) is  $140^\circ$       D) is  $120^\circ$       E) cannot be found

27. The ratio of the area of an equilateral triangle inscribed in a circle to that of a regular hexagon inscribed in the same circle is

- A) 1:2      B) 1:3      C) 1:1      D) 2:1      E) 3:1

28. How many three digit numbers are there the sum of whose digits is odd?

- A) 500      B) 450      C) 451      D) 499      E) 501

29. A circular pizza is cut into pieces by 7 straight knife cuts. What is the maximum number of pieces possible?

- A) 14      B) 29      C) 35      D) 49      E) 128

30. Six straight lines are drawn in a plane with no two parallel and no three concurrent. The number of regions into which they divide the plane is

- A) 16      B) 20      C) 22      D) 24      E) 26